

— NETWORK & SECURITY ENGINEER

Hassan Ismail.

Firewalls, routing & VPN – built and verified on FortiGate.

I design and harden network infrastructure end to end — high-availability firewall clusters, SD-WAN edges, site-to-site IPsec tunnels, and segmented DMZ architectures — then prove every path with a policy trace and a CLI ping.

“Failure is a milestone for success.”

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SELECTED WORK

01 HA Cluster <-> SD-WAN Site-to-Site VPN

Two FortiGate sites joined over an encrypted tunnel

02 DMZ Segmentation

Three-legged firewall isolating public-facing services

● HA CLUSTER
SYNCHRONIZED

● IPSEC TUNNEL
UP - 0% LOSS

● SD-WAN
DUAL-WAN ACTIVE

● DMZ
SEGMENTED



ID: H. ISMAIL

● AVAILABLE

What's actually behind the work

Every item here is exercised somewhere in the two builds in this book.

CORE SKILLS

- Firewall Policy Design
- High Availability (HA) Clustering
- Site-to-Site IPsec VPN
- SD-WAN (Multi-WAN)
- Network Segmentation (DMZ)
- Static & Failsafe Routing
- CLI-Based Verification & Diagnostics

TOOLS & PLATFORMS

- FortiOS / FortiGate VM64
- FortiGate CLI Console
- Python (Intro + Intermediate)
- Network Fundamentals & Protocols

Services

Scope drawn directly from the two builds in this book.

01

Firewall & Policy Design

Rule sets built around least privilege - explicit deny where it matters, scoped ACCEPT rules everywhere else, all under UTM inspection.

Project 01 - 02

02

High Availability

Active-passive FortiGate clustering with unit priority and synchronization, so a single hardware failure doesn't take the network down.

Project 01

03

Site-to-Site VPN

IPsec tunnels between independent sites, with routing that prefers the tunnel and falls back to a blackhole route if it drops.

Project 01

04

SD-WAN

Multi-WAN edges grouped into a single zone, so outbound traffic keeps moving even if one link degrades.

Project 01

05

Network Segmentation

Zone-based design - LAN, DMZ, WAN - with firewall policy controlling what can reach what.

Project 02

06

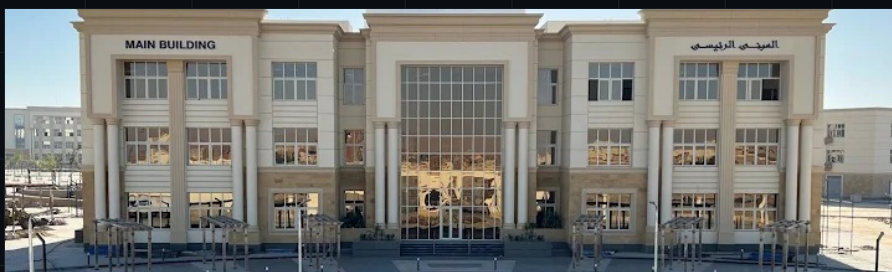
Verification & Testing

Every path is confirmed from the CLI - sourced pings, tunnel status, session counters - before it's called done.

Project 01 - 02

Currently building the fundamentals

Second-year Computer Science & AI student, with applied network-security practice and two completed Python courses alongside coursework.



ENTERING YEAR 3

South Valley National University

Faculty of Computer Science & Artificial Intelligence - CS & IT

Completed the second year, now entering the third. Coursework in programming, algorithms, and computer networks runs alongside the independent FortiGate work in this book.



NTI - 2026

National Telecommunication Institute

FortiOS Administrator -- Network Security, 60-hour track

Official Fortinet Academy curriculum: firewall policies, the Security Fabric, authentication, and security profiles (IPS, antivirus, web filtering, app control), plus IPsec VPN, SD-WAN, and HA -- the practical basis for the FortiGate projects in this book, built toward the NSE4 FortiOS Administrator path (exam NSE4_FGT_AD-7.6). Also completed two Microsoft-backed Python courses under an MCIT digital-skills program.



Intro to Python
Sep 14-23, 2025



Python - Intermediate
Sep 25-Oct 2, 2025

HA Cluster <-> SD-WAN Site-to-Site VPN

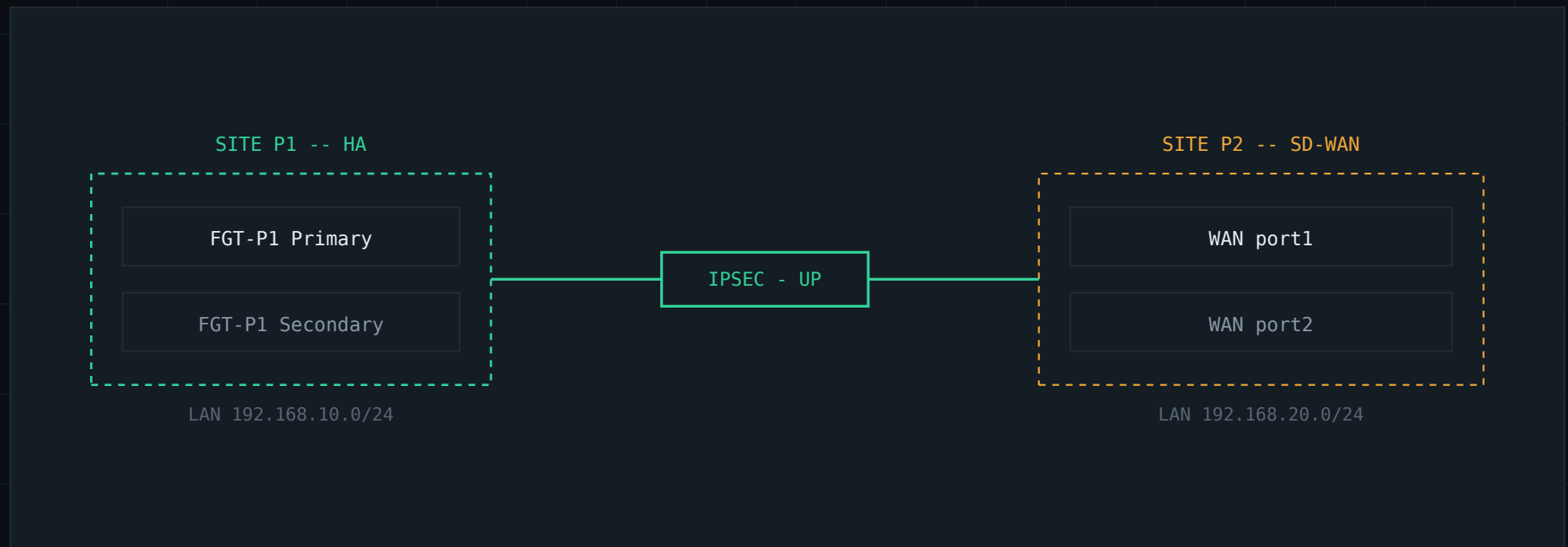
Two independent FortiGate deployments, built and validated as a single connected system: an active-passive HA pair at Site P1, and a dual-WAN SD-WAN edge at Site P2, joined by an IPsec tunnel and reachable subnet-to-subnet. Static routes on both sides prefer the tunnel and fall back to a blackhole route so traffic never leaks to the internet if the VPN drops.

Role: Primary / Secondary

Priority: 250

Tunnel: Up - both directions

Verification: CLI ping, 0% loss



Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Administrators

Admin Profiles

Firmware

Fabric Management

Settings

HA

SNMP

Replacement Messages

FortiGuard

Feature Visibility

Certificates

Security Fabric

Log & Report

HA: Primary

admin

FortiGate VM64

FortiGate VM64

FortiGate-VM64-HA-backup (Primary)

FortiGate-VM64 (Secondary)

Refresh

Edit

Remove device from HA cluster

Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
Synchronized	250	FortiGate-VM64-H	FGVMEVJZDR22AM0A	Primary	1m 15s	42	78.00 kbps
Synchronized	250	FortiGate-VM64	FGVMEVJWGIS_7T31	Secondary	3m 32s	27	22.00 kbps

01 HA Status

Active-passive HA pair, both synchronized at priority 250 - live session count and throughput per unit.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref
802.3ad Aggregate 1							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254	2
Physical Interface 11							
port1	Physical Interface		172.21.111.113/255.255.240.0	PING HTTPS SSH HTTP FMG-Access			4
port2	Physical Interface		0.0.0.0/0.0.0.0				0
port3	Physical Interface		192.168.10.1/255.255.255.0	PING HTTPS SSH			3
port4	Physical Interface		0.0.0.0/0.0.0.0				0

0 Security Rating Issues

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02 Interfaces

port1 carries WAN/management (SSH, HTTPS, FMG-Access); port3 terminates the internal 192.168.10.0/24 LAN.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

- Interfaces
- DNS
- Packet Capture
- SD-WAN
- Static Routes
- Policy Routes
- RIP
- OSPF
- BGP
- Routing Objects
- Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

HA: Primary

Create New

Edit

Clone

Delete

Search

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0	Dynamic Gateway (172.21.96.1)	port1	Enabled	
vpn to p2_remote		vpn to p2	Enabled	VPN: vpn to p2 (Created by VPN wizard)
vpn to p2_remote		Blackhole	Enabled	VPN: vpn to p2 (Created by VPN wizard)

3

03 Static Routing

Default route out port1, a primary route through the tunnel, and a blackhole route so nothing leaks if the VPN drops.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

HA: Primary

Create New

Edit

Delete

Policy Lookup

Search

Export

Interface Pair View

By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
port3 → port1 1									
LAN to internet	all	all	always	ALL	ACCEPT		no-inspection	UTM	0 B
port3 → vpn to p2 1									
vpn to p2 → port3 1									
Implicit 1									

0 Security Rating Issues

4 Updated: 09:37:18

04 Firewall Policy

LAN-to-internet plus dedicated inbound/outbound rules for the VPN interface, all under UTM inspection.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

HA: Primary

>

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admin

+ Create New

Edit

Delete

Search

Tunnel	Interface Binding	Status	Ref.
Site to Site - FortiGate 1			
vpn to p2	port1	Up	4

FORTINET v7.0.15

0 Security Rating Issues

1

05 IPsec Tunnel

Site-to-site tunnel "vpn to p2," bound to port1, status Up.

● HASSAN ISMAIL

● SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

The screenshot shows the FortiGate VM64-HA-backup CLI console. The left sidebar contains navigation options: Dashboard, Network, Policy & Objects, Security Profiles, VPN, Overlay Controller, IPsec Tunnels, IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System, Security Fabric, and Log & Report. The main console area displays the following commands and output:

```
FortiGate-VM64-HA-backup # execute ping-options source 192.168.10.1
FortiGate-VM64-HA-backup # execute ping 192.168.20.1
PING 192.168.20.1 (192.168.20.1): 56 data bytes
64 bytes from 192.168.20.1: icmp_seq=0 ttl=255 time=2.4 ms
64 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=1.2 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=1.7 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=1.5 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=1.6 ms

--- 192.168.20.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.2/1.6/2.4 ms

FortiGate-VM64-HA-backup #
```

The bottom status bar shows the Fortinet logo, version v7.0.15, and a security rating of 0. The user 'admin' is logged in.

06 CLI Verification

Sourced ping from the P1 LAN to the P2 LAN (192.168.20.1) - 5/5 packets, 0% loss, 1.6 ms average.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate 1							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection			1
Physical Interface 11							
port1	Physical Interface		172.21.109.73/255.255.240.0	PING HTTPS SSH HTTP FMG-Access			3
port2	Physical Interface		172.21.98.128/255.255.240.0	PING HTTPS SSH			1
port3	Physical Interface		192.168.20.1/255.255.255.0	PING HTTPS SSH			3
port4	Physical Interface		0.0.0.0/0.0.0.0				0

0 Security Rating Issues

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01 Interfaces

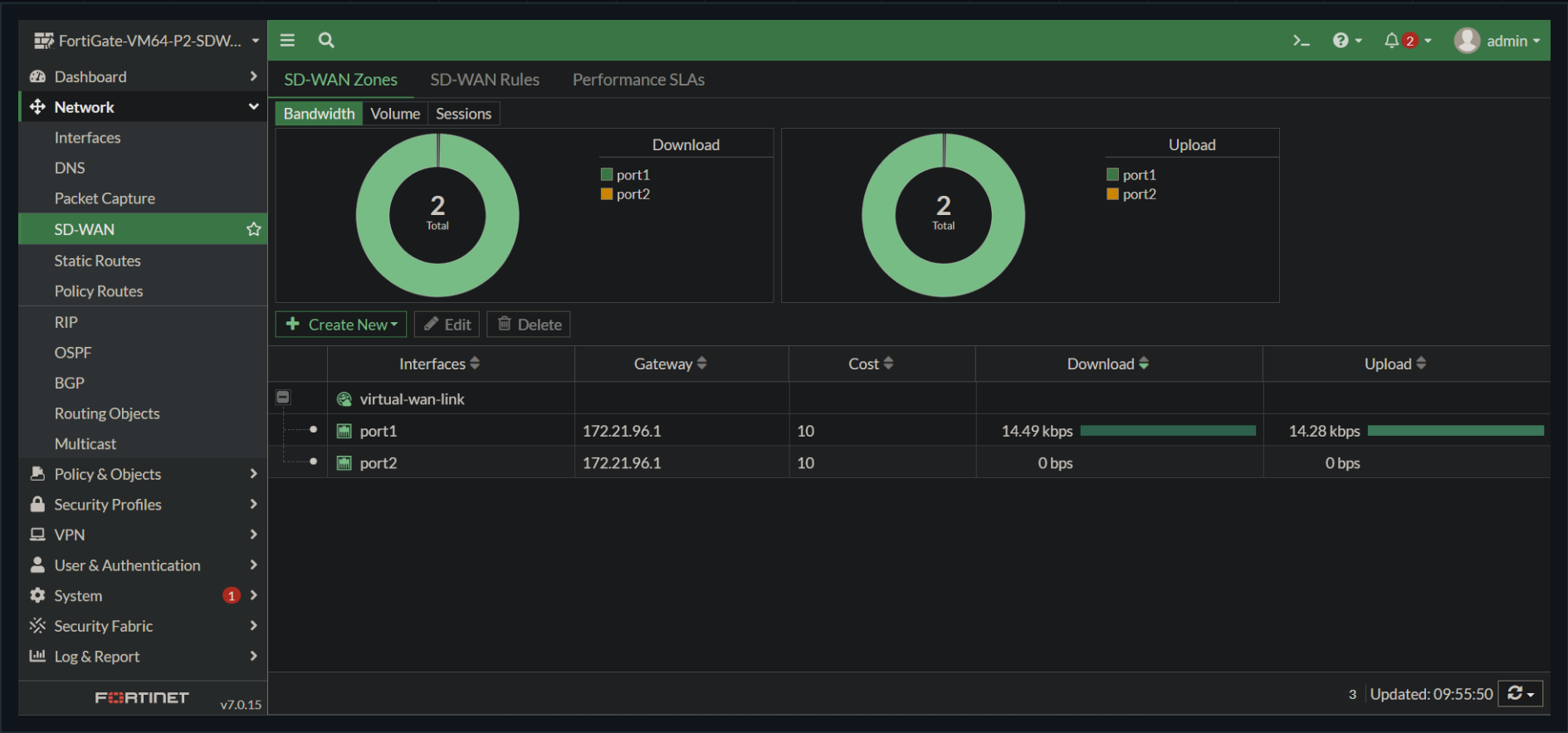
Two independent WAN links (port1, port2) plus the internal 192.168.20.0/24 LAN on port3.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up



02 SD-WAN Zone

Both WAN ports grouped into the virtual-wan-link at equal cost, with live download/upload throughput per member.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...

Dashboard

Network

- Interfaces
- DNS
- Packet Capture
- SD-WAN
- Static Routes
- Policy Routes
- RIP
- OSPF
- BGP
- Routing Objects
- Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FORTINET v7.0.15

Create New Edit Clone Delete Search

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0		virtual-wan-link	Enabled	
vpn to p1_remote		vpn to p1	Enabled	VPN: vpn to p1 (Created by VPN wizard)
vpn to p1_remote		Blackhole	Enabled	VPN: vpn to p1 (Created by VPN wizard)

3

03 Static Routing

Default route rides the SD-WAN zone; the route to P1 gets the same primary-plus-blackhole failsafe pattern as P1.

WAN links: port1 + port2

```
SD-WAN zone: virtual-wan-link
```

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiNET

v7.0.15

Create New

Edit

Delete

Policy Lookup

Search

Export

Interface Pair View

By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
port3 → virtual-wan-link 1									
lan to sdwan	all	all	always	ALL	ACCEPT	Enabled	SSL no-inspection	UTM	0 B
port3 → vpn to p1 1									
vpn to p1 → port3 1									
Implicit 1									

0 Security Rating Issues

4 Updated: 09:43:09

04 Firewall Policy

LAN-to-SD-WAN policy for general egress, mirrored by inbound/outbound rules for the tunnel back to P1.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...
Dashboard
Network
Policy & Objects
Security Profiles
VPN
Overlay Controller VPN
IPsec Tunnels
IPsec Wizard
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report

FortiGate-VM64-P2-SDW...
+ Create New
Edit
Delete
Search

Tunnel	Interface Binding	Status	Ref.
Site to Site - FortiGate 1			
vpn to p1	port1	Up	4

0 Security Rating Issues

05 IPsec Tunnel

Site-to-site tunnel "vpn to p1," bound to port1, status Up.

● HASSAN ISMAIL

● SITE P2 -- SD-WAN

PROJECT 01

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

The screenshot shows the FortiGate VM64-P2-SDWAN CLI Console. The left sidebar contains a navigation menu with options: Dashboard, Network, Policy & Objects, Security Profiles, VPN, Overlay Controller, IPsec Tunnels, IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System, Security Fabric, and Log & Report. The main console area displays the following output:

```
FortiGate-VM64-P2-SDWAN # execute ping-options source 192.168.20.1

FortiGate-VM64-P2-SDWAN # execute ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1): 56 data bytes
64 bytes from 192.168.10.1: icmp_seq=0 ttl=255 time=1.5 ms
64 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=0.9 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=1.5 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=0.9 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=1.2 ms

--- 192.168.10.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.9/1.2/1.5 ms

FortiGate-VM64-P2-SDWAN #
```

The bottom of the console shows the FortiGate logo, version 7.0.15, and a security rating of 0. The user 'admin' is logged in.

06 CLI Verification

Sourced ping from the P2 LAN to the P1 LAN (192.168.10.1) - 5/5 packets, 0% loss, 1.2 ms average, confirming bidirectional reachability.

DMZ Segmentation

A single FortiGate split into three security zones, so a compromised public-facing host in the DMZ can never pivot into the internal LAN. port1 faces the WAN, port2 hosts the internal LAN (192.168.10.0/24), and port3 is a dedicated DMZ (192.168.20.0/24) for exposed services. Policy does the isolating: DMZ-to-LAN traffic is explicitly denied, LAN-to-DMZ and LAN-to-WAN are permitted, and the implicit deny rule closes everything not named.

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

LAN -> WAN: Accepted



deny

allow

POLICY ORDER

- 1 dmz -> lan DENY
- 2 lan -> dmz ACCEPT
- 3 lan -> wan ACCEPT
- 4 any -> any DENY

implicit deny closes the rest

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

FortiGate-VM64

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate 1							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection			1
Physical Interface 10							
DMZ zone (port3)	Physical Interface		192.168.20.1/255.255.255.0	PING HTTPS SSH			3
LAN network (port2)	Physical Interface		192.168.10.1/255.255.255.0	PING HTTPS SSH			4
port1	Physical Interface		172.28.141.165/255.255.240.0	PING HTTPS SSH HTTP FMG-Access			1

0 SecurityRating Issues

0% 12 Updated: 09:07:46

01 Interfaces

Dedicated DMZ (port3), internal LAN (port2), and WAN (port1) on a single FortiGate - each its own subnet.

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

FortiGate-VM64

Dashboard
Network
Policy & Objects
Firewall Policy
IPv4 DoS Policy
Addresses
Internet Service Database
Services
Schedules
Virtual IPs
IP Pools
Protocol Options
Traffic Shaping
Security Profiles
VPN
User & Authentication
System
Security Fabric
Log & Report

FortiNET v7.0.15
0 Security Rating Issues
Updated: 09:07:56

admin

+ Create New
Edit
Delete
Policy Lookup

Export
Interface Pair View
By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
DMZ zone (port3) → LAN network (port2)									
block dmz to lan	dmz network	lan network	always	ALL	DENY			All	0 B
LAN network (port2) → DMZ zone (port3)									
lan to dmz access	lan network	dmz network	always	ALL	ACCEPT	Disabled	SSL no-inspection	All	0 B
LAN network (port2) → port1									
lan to wan	lan network	all	always	ALL	ACCEPT	Enabled	SSL no-inspection	All	0 B
Implicit									
Implicit Deny	all	all	always	ALL	DENY			Enabled	0 B

02 Firewall Policy

DMZ-to-LAN explicitly denied, LAN-to-DMZ and LAN-to-WAN permitted, implicit deny closes everything else.

● HASSAN ISMAIL

— GET IN TOUCH

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Qena, Egypt

● HA CLUSTER
SYNCHRONIZED

● IPSEC TUNNEL
UP - 0% LOSS

● SD-WAN
DUAL-WAN ACTIVE

● DMZ
SEGMENTED